

Network Automation and AI

Network automation is becoming an important part of modern IT infrastructure. Automation helps engineers manage large networks efficiently and reduces manual errors.

Common network automation tools include:

1. Python
2. Ansible
3. Netmiko
4. Nornir
5. Terraform

Python is widely used because it has strong networking libraries and is easy to learn.

Engineers often automate tasks like:

- configuration backups
- device health checks
- firmware upgrades
- network inventory collection

Using AI in Network Operations

Artificial Intelligence is transforming network operations centers (NOC).

AI can help with:

- anomaly detection
- root cause analysis
- automated troubleshooting
- predictive maintenance

For example, an AI system can analyze logs from routers and switches and predict failures before they occur.

This helps reduce downtime and improves service reliability.

RAG for Network Knowledge

Retrieval Augmented Generation (RAG) allows AI systems to retrieve relevant documents before generating answers.

A typical RAG pipeline includes:

1. Document Loading
2. Text Chunking
3. Embeddings
4. Vector Database
5. Retrieval
6. LLM Response Generation

LangChain provides tools like TextLoader, PyPDFLoader, and RecursiveCharacterTextSplitter to build RAG pipelines.